

SUPERSTORE SALES ANALYSIS DASHBOARD

End-to-End Data Analytics Project

Using

PostgreSQL | SQL | Power BI | DAX

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PROJECT OVERVIEW

Introduction

The Superstore Sales Analysis Dashboard is an end-to-end Data Analytics project developed to analyze sales performance using PostgreSQL, SQL, Power BI, and DAX.

The project focuses on transforming raw sales data into meaningful business insights through SQL analysis and interactive Power BI dashboards.

Project Purpose

The main purpose of this project is to help businesses understand their sales performance, identify profitable products and regions, monitor key performance indicators (KPIs), and support better business decisions using data visualization.

Project Workflow

CSV Dataset



PostgreSQL Database



SQL Analysis



Power BI



DAX Measures



Interactive Dashboard



Business Insights

PROJECT OBJECTIVES

Objectives

The primary objectives of this project are:

- Analyze overall sales performance.
- Measure total profit and sales growth.
- Identify the best-performing regions.
- Compare sales across different product categories.
- Identify top-selling products.
- Analyze customer purchasing behavior.
- Monitor monthly sales trends.
- Create interactive dashboards for business users.
- Support data-driven decision making.
- Improve reporting using Power BI and DAX.

Expected Outcome

This project provides meaningful business insights through interactive dashboards and SQL analysis. It enables decision-makers to monitor business performance, identify profitable opportunities, and improve strategic planning using data.

DATASET DESCRIPTION

Dataset Overview

This project uses the Sample Superstore dataset, which contains sales transactions of a retail business. The dataset includes customer information, product details, sales, profit, discount, quantity, shipping details, and geographical information.

The dataset is widely used for data analytics, SQL practice, and Power BI dashboard development.

Dataset Details

Property	Value
Dataset Name	Sample Superstore
File Format	CSV
Database	PostgreSQL
Visualization Tool	Power BI
Analysis Language	SQL
Calculated Measures	DAX

Important Columns

- Order ID
- Order Date
- Customer Name
- Region
- Category
- Sub-Category
- Product Name
- Sales
- Quantity
- Profit

POSTGRESQL DATABASE

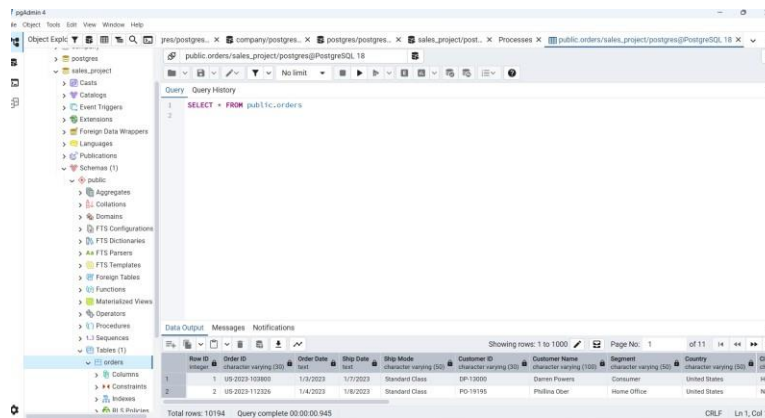
Database Setup

The Sample Superstore dataset was imported into PostgreSQL to perform SQL-based analysis. A dedicated database and table were created, and the CSV file was loaded successfully for querying and reporting.

Implementation Steps

- Created a PostgreSQL database.
- Created the Orders table using SQL.
- Imported the Sample Superstore CSV dataset.
- Verified the imported records.
- Checked data types and column names.
- Connected PostgreSQL with Power BI.

Database Screenshot



The screenshot shows the PostgreSQL pgAdmin 4 interface. On the left, the 'Object Explorer' pane displays the database structure, including the 'public' schema and the 'orders' table. The 'Query Editor' pane shows a SQL query: `SELECT * FROM public.orders`. The 'Data Output' pane displays the first two rows of the 'orders' table. The table has 11 columns: Row ID, Order ID, Order Date, Ship Date, Ship Mode, Customer ID, Customer Name, Segment, Country, and City. The first row shows an order with ID 1, dated 1/5/2023, shipped via 'Standard Class' to customer 'GP-13050' in the 'Consumer' segment, located in the 'United States'.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
1	109-2023-103800	1/5/2023	1/7/2023	Standard Class	GP-13050	Darren Powers	Consumer	United States	Nap
2	109-2023-112326	1/4/2023	1/6/2023	Standard Class	PO-19195	Phillips Oser	Home Office	United States	Nap

Total rows: 10194 Query complete 00:00:00.945

Figure 1: PostgreSQL Orders Table

Superstore Sales Analysis Dashboard

SQL ANALYSIS

Project SQL Analysis

SQL was used to analyze the Superstore dataset and answer important business questions. Multiple queries were written to evaluate sales performance, profitability, customer behavior, and regional performance.

Business Questions Solved

- Calculate Total Sales
- Calculate Total Profit
- Count Total Orders
- Find Top 10 Customers
- Find Top 10 Products
- Region-wise Sales Analysis
- Category-wise Sales Analysis
- Monthly Sales Trend
- State-wise Profit Analysis
- Average Order Value

Key Learning

This phase of the project improved SQL skills by using aggregate functions, GROUP BY, ORDER BY, filtering, and analytical queries to extract meaningful business insights from the dataset.

SQL Screenshot

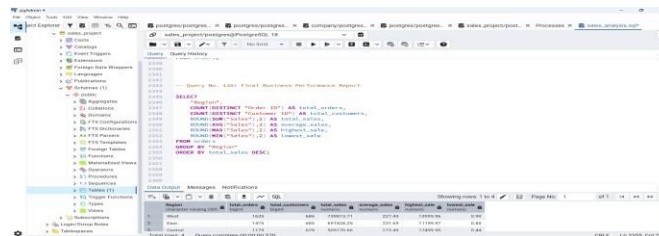


Figure 1: PostgreSQL Database

POWER BI DASHBOARD

Dashboard Overview

An interactive Power BI dashboard was developed to visualize sales performance and business metrics. The dashboard enables users to analyze sales trends, profit, customer behavior, and regional performance through interactive charts, KPI cards, and slicers.

Dashboard Components

- KPI Cards (Sales, Profit, Orders, Customers)
- Monthly Sales Trend
- Region-wise Sales Analysis
- Category-wise Sales Analysis
- Interactive Slicers
- Navigation Buttons
- Order Details Table

Dashboard Screenshot



Figure 3: Power BI Dashboard

Key Benefits

- Easy to understand business performance.
- Interactive filtering for better analysis.
- Quick identification of profitable regions.
- Supports data-driven decision making.

DETAILED DASHBOARD ANALYSIS

Dashboard Analysis

The second dashboard provides a detailed analysis of sales performance by focusing on customers, products, cities, and order-level information. It helps users perform in-depth analysis and identify high-performing business areas.

Dashboard Components

- Top 10 Customers
- Top 10 Products
- Top 10 Cities
- Order Details Table
- Dynamic Filters
- Interactive Navigation

Detailed Dashboard Screenshot

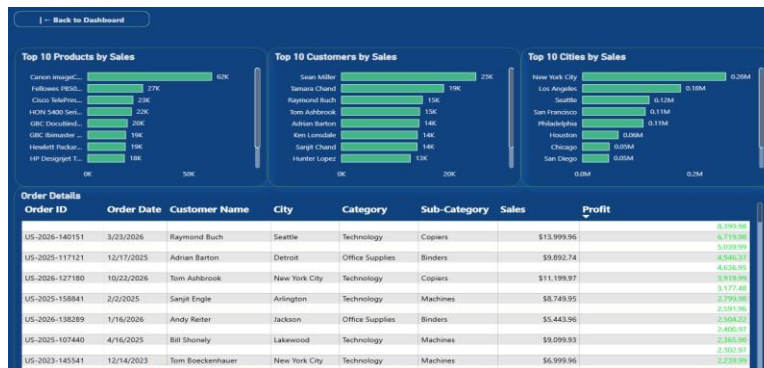


Figure 4: Detailed Dashboard Analysis

Business Value

This dashboard enables businesses to identify their most valuable customers, best-selling products, and high-performing cities. It also provides order-level insights that support better operational and strategic decisions.

DAX MEASURES

Introduction to DAX

Data Analysis Expressions (DAX) were used in Power BI to create calculated measures for analyzing sales performance. These measures provide dynamic calculations that automatically update based on filters and slicers applied in the dashboard.

DAX Measures Created

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Average Sales
- Profit Margin

Purpose of DAX

DAX measures improve the dashboard by performing real-time calculations and providing accurate business metrics. They allow users to analyze sales performance dynamically based on selected filters, regions, categories, and time periods.

Sample DAX Measures

Total Sales = SUM(Orders[Sales])

Total Profit = SUM(Orders[Profit])

Total Orders = DISTINCTCOUNT(Orders[Order ID])

BUSINESS INSIGHTS

Key Business Insights

- Sales performance varies across different regions.
- Some product categories generate higher profits than others.
- Monthly sales trends help identify seasonal business patterns.
- A small group of customers contributes significantly to total sales.
- Top-selling products generate a major share of revenue.
- Interactive dashboards make business analysis faster and easier.
- Data-driven insights help improve strategic business decisions.

Project Outcomes

This project demonstrates how raw business data can be transformed into meaningful insights using PostgreSQL, SQL, Power BI, and DAX. The interactive dashboard helps stakeholders monitor business performance, identify opportunities, and make informed decisions.

Skills Applied

- SQL Query Writing
- Database Management
- Data Cleaning
- Data Analysis
- Power BI Dashboard Development
- DAX Measures
- Data Visualization
- Business Intelligence
- Monthly sales trends help identify seasonal business patterns.
- A small group of customers contributes significantly to total sales.
- Top-selling products generate a major share of revenue.
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Skills Applied

- SQL Query Writing
- Database Management
- Data Cleaning
- Data Analysis
- Power BI Dashboard Development
- DAX Measures

CONCLUSION

Project Summary

This project successfully demonstrates an end-to-end data analytics workflow using PostgreSQL, SQL, Power BI, and DAX. The Superstore Sales Analysis Dashboard transforms raw sales data into meaningful business insights through interactive visualizations and analytical reporting.

Key Achievements

- Built a PostgreSQL database for data storage.
- Performed business analysis using SQL queries.
- Connected PostgreSQL with Power BI.
- Created DAX measures for dynamic calculations.
- Designed an interactive sales dashboard.
- Generated valuable business insights.
- Completed an end-to-end data analytics project.

Future Improvements

- Add sales forecasting.
- Integrate live database connectivity.
- Include customer segmentation analysis.
- Develop advanced KPI dashboards.
- Automate report refresh.

Final Remark

This project strengthened my practical knowledge of SQL, PostgreSQL, Power BI, DAX, and business intelligence. It reflects my ability to build complete data analytics solutions and present insights in a professional manner.

Superstore Sales Analysis Dashboard